

Coding & Solution

Date	3 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation details of the Smart Lender project. The application is developed using Python, Flask, and Machine Learning to predict loan eligibility. The codebase follows

a modular structure with separate components for data preprocessing, model training, prediction, and the web interface, ensuring maintainability and scalability.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mohithimandi/Smart-Lender.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	Loan eligibility prediction, data preprocessing, multiple ML models, XGBoost model selection, Flask web interface, real-time prediction, Hugging Face deployment
Pending / Incomplete Features	User authentication, prediction history, database integration (<i>if not implemented</i>)
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , run <code>python app.py</code> , and open the application in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The SmartLender project follows clean coding practices with a modular architecture that separates the frontend, backend, and machine learning components. The application uses the XGBoost model for accurate loan eligibility prediction and is deployed on Hugging Face for easy online access. The code is organized, reusable, and designed to support future enhancements such as user authentication, database integration, and prediction history.